Daily Drilling Report
SPD19B-03
9/9/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
44.0	8 1/2"	189.00	72.83	31.70	91.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	264,057.41	4,484.23	10,835,324.00		39.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,949.00	3,039.00	2,864.54	90.00	9.60	9.4

Operations at Report Time
Drilling @ 3113m.

Operations Summary
RIH directional BHA#11 to bottom at 2949m. Fan the bottom & profile the bit.Resume directional drilling 8 1/2" hole section f/ 2949m to 3039m.

Operations next 24hrs
Cont' drilling 8 1/2" hole section. Sweep the hole. Perform wiper trip.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/9/2014	10/25/2014
Trip Drill	9/8/2014	10/29/2014

STOP Cards

Number of Reports: 36

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	01:30	1.50	8 1/2"	D	04	P	P/U, M/U 8 1/2" directional BHA#11 consisting of items below and RIH to 22m. Align Motor with MWD. 1) 8 1/2" New Varel PDC bit (Type: VM713GUX, IADC: M332, SN: 6006868, w/ 4 x 14/32" nozzles) 2) 6 3/4" Mud Motor (Bent housing: 1.25°, Sleeve Stabilizer: 8 3/8" , Rev./gal: 0.28 , Lobes: 7/8 , Stage:5) 3) used 8" COR Stabilizer (SN: COR-2941-AD) 4) 6 3/4" NMDC 5) 6 3/4" MWD TMS
01:30	02:00	0.50	8 1/2"	D	12	P	Connect TDS & perform MWD & Motor shallow hole test w/ 500 gpm/ 1100 psi, ok.

02:00	04:00	2.00	8 1/2"	D	04	P	P/U the rest of BHA#11 consisting of items below and RIH to 252m.
							
							06) 6 3/4" NMDC
							07) 6 3/4 F. Sub
							08) XOS (4 1/2" IF pin x XT57 box)
							19) 20 Jts 5 7/8" HWDP
							10) used 7" drilling jar (3021-AD)
							11) 3 Jts 5 7/8" HWDP
							
							>>> Weight below jar (in mud): 38 kips
							>>> Total BHA weight (in mud): 45 kips

04:00	09:30	5.50	8 1/2"	D	04	P	RIH directional drilling BHA#11 on 5 7/8" DP f/ 252m to 2719m.
09:30	10:30	1.00	8 1/2"	D	24	N	(NPT for ABAN) Observe leakage from Remote IBOP crank while POOH. Stop POOH at shoe. Change out upper IBOP and saver sub. Meanwhile monitor the well on T.Tank.
10:30	12:00	1.50	8 1/2"	D	04	P	Cont' RIH directional BHA#11 on 5 7/8" DP f/ 2719m to 2921m. W/down f/ 2921m w/ 400 gpm/ 1560psi and tag bottom @ 2949m.
12:00	00:00	12....	8 1/2"	D	03	P	Pump 20 bbl Hi-Vis pill and fan bottom w/ 500 gpm. Profile new bit. Resume directional drilling 8 1/2" hole f/ 2949m to 3039m w/ 500-550 gpm/ 2150-2500 psi/ 5-25 kips/ 30-60 rpm/ 17-22 TRQ. * Drilling with NL/NG. * Pump 20 bbl Hi-Vis pill when STD down. * B/R and R/D each STD drilled once. * Take survey before connection.

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		8 1/2"				Sep 10, 2014 00:00 - 06:00 Cont' directional drilling 8 1/2" hole in rotation mode f/ 3039m to 3113m w/ 550 gpm/ 2300-2600 psi/ 5-25 kips/ 60 rpm/ 18-25 TRQ. * Drilling with NL/NG. * Pump 30 bbl Hi-Vis pill when STD down. * B/R and R/D each STD drilled once. * Take survey before connection. >> Last sample @ 3095m: Lst. (100 %)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
3,039.00	CaCo3/KCL/ Polymer	10.50	54	26.0	20.000	2.7
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.8	0.600	1.000	1.200			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	7	0.0	85.0	15.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	8.000	10.000	0/100	61.0	1508.0	2825.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7576.0	50.0	0.0	1.90		12.81472	0.423

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
2.00	6.	315	22.0	999.0	37.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Ahmed 6	Supply Boat	9/7/2014 17:30		B/Loaded Cargo as Storage, Streaming at Location.
Samarghand	Supply Boat	9/9/2014 17:45	9/10/2014 00:10	Delivered Cargo. Received 40 MT FO from OSV NAVIS KING. departed for Sahar-1.
Navis King	Supply Boat	9/9/2014 20:00	9/9/2014 23:55	Delivered all manifested cargo. departed for Aban-8.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Caustic Soda 2014	15.0	500.0	287.0
Defoamer	1.0	8.0	5.0
Duo-Tec	3.0	185.0	110.0
KCL	1.0	52.0	18.0
Pac-LV	5.0	521.0	73.0
Starch	20.0	1,600.0	357.0
Soda Ash 25 Kg	35.0	750.0	272.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	21.0	10.0	175.0
Diesel Fuel	8.0	0.0	154.0
Drill Water	10.0	0.0	304.0
Cement G	0.0	0.0	92.0
Barite Bulk	0.0	0.0	129.0
Barite BB	0.0	0.0	0.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
11	8 1/2	Varel	VM713GUX	6006868	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	9.60	121,732.80	2,949.00	3,039.00	90.00	9.4	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	550	2,150.0	2,450.0	5	25	170	214
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
12.22	9.60	100.00	0.00	456	371	301	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,949.00	2,950.00	0.58	18,000.0	500	30	2,150.0	5	
2,950.00	3,039.00	9.02	19,000.0	550	60	2,450.0	25	371

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWPD	5 7/8	4.000	27.89	252.82
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	224.93
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	219.13
X/O	6 1/2	2.813	0.85	32.78
6 3/4" Float sub	6 3/4		0.81	31.93
6 3/4" NMDC	6 3/4	3.250	9.48	31.12

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
6 3/4" TM - Sub	6 3/4		1.74	21.64
6 3/4" NMDC	6 3/4	3.250	9.48	19.90
8" St. Stabilizer (COR-2941AD)	8	3.000	2.14	10.42
6 3/4" Motor (AKO: 1.25° , Sleeve: 8 3/8")	6 3/4		8.00	8.28
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
1	2,977.00	14.00	20	290.0	97	114
1	2,977.00	14.00	30	420.0	97	170
1	2,977.00	14.00	40	590.0	97	227
2	2,977.00	14.00	20	290.0	97	114
2	2,977.00	14.00	30	420.0	97	170
2	2,977.00	14.00	40	590.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,960.50	251.90	26.40	2,794.21	-372.65	-839.92	918.86	0.19	0.09
2,989.68	251.00	26.40	2,820.34	-376.78	-852.22	931.79	0.41	0.00
3,018.81	250.40	26.30	2,846.45	-381.05	-864.42	944.68	0.29	-0.10
3,048.15	249.80	26.40	2,872.74	-385.49	-876.66	957.67	0.29	0.10
3,077.42	249.60	26.40	2,898.95	-390.00	-888.87	970.67	0.09	0.00
3,106.10	249.10	26.50	2,924.63	-394.51	-900.82	983.42	0.26	0.10

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Upper Kangan		2,703.00	2,559.14
K1		2,732.00	2,586.77
K2		2,851.80	2,696.47
K3		2,901.00	2,740.86
K4		3,029.50	2,856.03

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14)

>>> Sent:

** Nil

>>> Received:

** Samarghand f/ Kish: (81 jts 7" 28% CR 23 PPF TBG) + (54 jts 7" 28% CR 26 PPF) + (2 x Liner Hanger) + (Mehran Completion BOX)

** Navis King f/ Sharjah: (ABAN 5 7/8" D.Pipes) + (food provision) + (Alghaith 7 5/8" Power Tong & Accessories) + (CTR 1 x 7" scrapper + 1 x 9 5/8" scrapper + 2 x 4 1/2" IF pin x 4 1/2 Reg pin XOver + 2 x 7.2" Polish Mill + 2 x CMT manifolds & related cross overs + 6 x 7" Landing collar + 3 x 5.875" Pump down plug + 6 x 7" wiper plug)